

Lab 1

Web Architecture

Tasnif Emran

220041135

MVC

Q1: When the user selects option 3 (Check out a book) and enters a valid ISBN, trace the exact sequence of method calls starting from `LibraryController.run()`. List each method call in order and state which component (Model, View, or Controller) it belongs to.

Ans:

From `LibraryController.run()` the call sequence for option 3 with a valid, available ISBN is:

`LibraryController.run()` (Controller)

`LibraryView.showMenu()` (View)

`LibraryView.readInt()` (View)

`LibraryController.handleCheckOut()` (Controller)

`LibraryView.askIsbn()` (View)

`Library.findByIsbn(isbn)` (Model)

`Book.isCheckedOut()` (Model)

`Book.checkOut()` (Model)

`LibraryView.showMessage()` (View)

`Book.getTitle()` (Model)

Q2: In `LibraryView.showBookList()`, the method receives a `List<Book>` parameter. Why does the View receive data as a parameter instead of calling `model.getAllBooks()` directly? What MVC principle does this enforce?

Ans: To make the view independent from the model.

Q3: The `getStatus()` method is defined in `Book.java` (the Model). Could this method be placed in `LibraryView.java` instead? Explain why or why not, considering MVC responsibilities.

Ans: `Book.getStatus()` belongs in the Model because it expresses business state. Putting it in the View would mix responsibilities.

Q4: If you wanted to replace the console interface with a graphical (GUI) interface, which classes would you need to change? Which classes would remain completely untouched? Explain why.

Ans: View changed. Model Untouched. And the Controller can remain unchanged as long as the new View provides the same interaction methods.

Q5: Draw a sequence diagram (on paper or using a diagramming tool) showing the complete interaction between Main, Controller, View, and Model when the user selects option 3 and enters the ISBN "978-0132350884" for a book that is currently available.

Ans:

actor User

participant Main

participant Controller as LibraryController

participant View as LibraryView

participant Model as Library/Book

Main->>Model: new Library()

Main->>View: new LibraryView()

Main->>Controller: new LibraryController(model, view)

Main->>Controller: run()

User->>View: selects option 3
Controller->>View: showMenu()
View->>View: readInt()
View-->>Controller: choice=3

Controller->>Controller: handleCheckOut()
Controller->>View: askIsbn()
User->>View: enters "978-0132350884"
View-->>Controller: isbn

Controller->>Model: findByIsbn(isbn)
Model-->>Controller: Book

Controller->>Model: isCheckedOut()
Model-->>Controller: false (available)

Controller->>Model: checkOut()
Controller->>Model: getTitle()
Model-->>Controller: "Clean Code"

Controller->>View: showMessage("Successfully checked out: Clean Code")